

Lynbrook Math Summer Camp 2026

N3 GCD and LCM

Lynbrook Math Club

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N3.1 Lecture

Suppose three bells ring once every 12, 18, and 30 minutes. If all three bells ring together now, when will they ring together again?

Or suppose Lynbrook Math Summer Camp has 72 6th graders, 108 7th graders, and 180 8th graders. The students are divided into identical classes each with the same number of students in each grade. What is the greatest possible number of classes?

These two problems appear different. The first asks when several repeating cycles will next coincide, while the second asks how to divide several quantities into the largest possible number of equal groups. However, both problems are connected by 2 closely related ideas: the **greatest common divisor** and the **least common multiple**.

Greatest Common Divisors

Recall that the notation

$$d \mid a$$

means that d divides a , or equivalently that a is divisible by d . In other words, there is some integer k such that

$$a = dk.$$

Definition N3.1.1 (Greatest Common Divisor). Let a and b be positive integers. The **greatest common divisor** of a and b , denoted

$$\gcd(a, b),$$

is the greatest positive integer d such that

$$d \mid a \quad \text{and} \quad d \mid b.$$

For example,

$$\gcd(24, 36) = 12$$

because 12 divides both 24 and 36, and no greater positive integer divides both.

Example 2 (Identical Bags)

A deck of cards has 72 red cards, 108 blue cards, and 180 green cards. The cards are divided into identical bags with no cards left over. What is the greatest possible number of bags? (This is just the camp problem, but with cards)

If there are d bags, then d must divide each of

$$72, \quad 108, \quad 180.$$

Therefore, the greatest possible number of bags is

$$\gcd(72, 108, 180).$$

We will soon learn several ways to calculate this number.

Least Common Multiples

Definition N3.1.3 (Least Common Multiple). Let a and b be positive integers. The **least common multiple** of a and b , denoted

$$\text{lcm}(a, b),$$

is the least positive integer m such that

$$a \mid m \quad \text{and} \quad b \mid m.$$

For example,

$$\text{lcm}(12, 18) = 36$$

because 36 is divisible by both 12 and 18, and no smaller positive integer has this property.

It is important that the definition asks for the least *positive* common multiple. Since 0 is divisible by every positive integer, allowing 0 would make every LCM equal to 0.

Example 4 (Ringing Bells)

Three bells ring once every 12, 18, and 30 minutes. If all three bells ring together at noon, when will they next ring together?

The number of minutes until the bells next ring together must be divisible by all three periods. Therefore, it is

$$\text{lcm}(12, 18, 30).$$

Prime Factorization Method

Recall from earlier that we can represent the prime factorizations of some 2 positive integers a and b as

$$a = \prod_p p^{\alpha_p} \quad \text{and} \quad b = \prod_p p^{\beta_p},$$

where the product ranges over all relevant primes with the power of prime p in a, b being α_p, β_p respectively (the notation for the prime factorization class from prior may have been slightly different, but overall it's the same idea).

A common divisor cannot contain more copies of a prime than either number. Therefore, for each prime, the GCD uses the **smaller exponent**.

Claim N3.1.5 (Prime-Exponent Formula for the GCD) —

$$\gcd(a, b) = \prod_p p^{\min(\alpha_p, \beta_p)}.$$

By similar logic, a common multiple must contain enough copies of every prime to be divisible by both numbers. Therefore, for each prime, the LCM uses the **larger exponent**.

Claim N3.1.6 (Prime-Exponent Formula for the LCM) —

$$\text{lcm}(a, b) = \prod_p p^{\max(\alpha_p, \beta_p)}.$$

Example 7

Find the GCD and LCM of 756 and 1260.

First factor:

$$756 = 2^2 \cdot 3^3 \cdot 7$$

and

$$1260 = 2^2 \cdot 3^2 \cdot 5 \cdot 7.$$

For the GCD, choose the smaller exponent of each prime:

$$\text{gcd}(756, 1260) = 2^2 \cdot 3^2 \cdot 7 = 252.$$

For the LCM, choose the larger exponent:

$$\text{lcm}(756, 1260) = 2^2 \cdot 3^3 \cdot 5 \cdot 7 = 3780.$$

We can now return to our opening problems.

For the cards,

$$72 = 2^3 \cdot 3^2, \quad 108 = 2^2 \cdot 3^3, \quad 180 = 2^2 \cdot 3^2 \cdot 5.$$

Thus,

$$\text{gcd}(72, 108, 180) = 2^2 \cdot 3^2 = 36.$$

The cards can be divided into at most 36 identical bags. Each bag contains

$$\frac{72}{36} = 2$$

red cards,

$$\frac{108}{36} = 3$$

blue cards, and

$$\frac{180}{36} = 5$$

green cards.

For the bells,

$$12 = 2^2 \cdot 3, \quad 18 = 2 \cdot 3^2, \quad 30 = 2 \cdot 3 \cdot 5.$$

Therefore,

$$\text{lcm}(12, 18, 30) = 2^2 \cdot 3^2 \cdot 5 = 180.$$

The bells next ring together after 180 minutes, or 3 hours. Thus, they next ring together at 3:00 PM.

The Euclidean Algorithm

Prime factorization works well when the numbers factor easily. However, factoring large numbers such as 756 and 1260 in our previous example may be difficult and annoying. The Euclidean algorithm finds the GCD without requiring the prime factorizations of the two numbers and sometimes provides a faster approach.

Claim N3.1.8 — For positive integers a, b and any integer q ,

$$\gcd(a, b) = \gcd(b, a - qb).$$

In particular, if

$$a = qb + r,$$

then

$$\gcd(a, b) = \gcd(b, r).$$

Proof. Suppose d divides both a and b . Since $d \mid a$ and $d \mid qb$, it also divides their difference:

$$a - qb.$$

Therefore, every common divisor of a and b is also a common divisor of b and $a - qb$.

Conversely, suppose d divides both b and $a - qb$. Then d divides

$$(a - qb) + qb = a.$$

Therefore, every common divisor of b and $a - qb$ is also a common divisor of a and b .

The two pairs have exactly the same common divisors, so they have the same greatest common divisor. $\square$

Example 9

Find

$$\gcd(2026, 646).$$

Repeatedly divide and keep the remainder:

$$2026 = 3(646) + 88,$$

so

$$\gcd(2026, 646) = \gcd(646, 88).$$

Next,

$$646 = 7(88) + 30,$$

so

$$\gcd(646, 88) = \gcd(88, 30).$$

Continue:

$$88 = 2(30) + 28,$$

$$30 = 28 + 2,$$

$$28 = 2(14) + 0.$$

The final nonzero remainder is 2, so

$$\gcd(2026, 646) = 2.$$

Properties of GCDs and LCMs

The GCD–LCM Product Identity

Claim N3.1.10 — For any positive integers a and b ,

$$\boxed{\gcd(a, b) \operatorname{lcm}(a, b) = ab}.$$

Proof. Consider any prime p . Suppose its exponent in a is r , and its exponent in b is s .

Its exponent in the GCD is

$$\min(r, s),$$

while its exponent in the LCM is

$$\max(r, s).$$

Therefore, its total exponent in the product of the GCD and LCM is

$$\min(r, s) + \max(r, s) = r + s.$$

But $r + s$ is exactly the exponent of p in ab . Since this is true for every prime,

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab.$$

□

Remark N3.1.11. Note: This identity applies to *two* positive integers. In general,

$$\gcd(a, b, c) \operatorname{lcm}(a, b, c) \neq abc.$$

For example,

$$\gcd(2, 4, 8) \operatorname{lcm}(2, 4, 8) = 2 \cdot 8 = 16,$$

whereas

$$2 \cdot 4 \cdot 8 = 64.$$

However, we do have

$$\gcd(a, b, c) \operatorname{lcm}(ab, bc, ca) = \operatorname{lcm}(a, b, c) \gcd(ab, bc, ca) = abc.$$

So, in this previous example,

$$\gcd(2, 4, 8) \operatorname{lcm}(2 \cdot 4, 4 \cdot 8, 8 \cdot 2) = 2 \cdot 32 = 64 = 2 \cdot 4 \cdot 8,$$

$$\operatorname{lcm}(2, 4, 8) \gcd(2 \cdot 4, 4 \cdot 8, 8 \cdot 2) = 8 \cdot 8 = 64 = 2 \cdot 4 \cdot 8.$$

Relatively Prime Numbers

Definition N3.1.12. Two positive integers a and b are called **relatively prime** or **coprime** if

$$\gcd(a, b) = 1.$$

The numbers themselves do not have to be prime. For example, 8 and 15 are relatively prime even though both are composite.

From the product identity, if $\gcd(a, b) = 1$, then

$$\boxed{\operatorname{lcm}(a, b) = ab}.$$

Example 13

Since

$$\gcd(8, 15) = 1,$$

we have

$$\text{lcm}(8, 15) = 8 \cdot 15 = 120.$$

However,

$$\text{lcm}(8, 12) \neq 8 \cdot 12$$

because 8 and 12 share a factor. In fact,

$$\text{lcm}(8, 12) = 24.$$

A Divisibility Consequence

Claim N3.1.14 — Suppose

$$a \mid n, \quad b \mid n, \quad \gcd(a, b) = 1.$$

Then

$$\boxed{ab \mid n}.$$

Proof. Because n is divisible by both a and b , it must be divisible by their least common multiple:

$$\text{lcm}(a, b) \mid n.$$

Since a and b are relatively prime,

$$\text{lcm}(a, b) = ab.$$

Therefore,

$$ab \mid n.$$

□

For example, if $8 \mid n$ and $15 \mid n$, then

$$120 \mid n.$$

But if $8 \mid n$ and $12 \mid n$, we may conclude only that

$$24 \mid n,$$

not necessarily that $96 \mid n$ (specifically, 72 contradicts this).

GCDs and Linear Combinations

If d divides both a and b , then d divides every integer linear combination

$$ax + by.$$

Indeed, if

$$a = dm \quad \text{and} \quad b = dn,$$

then

$$ax + by = d(mx + ny).$$

Therefore,

$$d \mid ax + by.$$

This is the main idea behind many contest problems involving GCDs of algebraic expressions.

Claim N3.1.15 (Bézout's Identity) — For any positive integers a and b , there exist integers x and y such that

$$ax + by = \gcd(a, b).$$

Bézout's identity may be proved by reversing the steps of the Euclidean algorithm. It is left as one of the exercises.

GCDs of Algebraic Expressions

The Euclidean algorithm works with expressions as well as ordinary numbers. We can also apply it on terms containing variables or use it to generalize relations for numbers, like below:

Claim N3.1.16 — Consecutive positive integers are relatively prime:

$$\gcd(n, n + 1) = 1.$$

Proof. Any common divisor of n and $n + 1$ must divide their difference:

$$(n + 1) - n = 1.$$

Therefore, their only positive common divisor is 1. □

More generally, any common divisor of n and $n + k$ divides

$$(n + k) - n = k.$$

Therefore,

$$\gcd(n, n + k) \mid k.$$

Example 17

What are the possible values of

$$\gcd(n, n + 12)?$$

The GCD must divide 12, so it must be one of

$$1, 2, 3, 4, 6, 12.$$

Each value is achievable. For example, choosing $n = d$, where $d \mid 12$, gives

$$\gcd(d, d + 12) = d.$$

Therefore, all possible values are

$$1, 2, 3, 4, 6, 12.$$

Example 18

Find

$$\gcd(3n + 2, 5n + 3).$$

Any common divisor divides every integer linear combination of the two expressions. In particular,

$$5(3n + 2) - 3(5n + 3) = 1.$$

Thus, every common divisor must divide 1. Therefore,

$$\boxed{\gcd(3n + 2, 5n + 3) = 1}.$$

Example 19

Find

$$\gcd(2n + 3, 4n + 8).$$

A common divisor must divide

$$(4n + 8) - 2(2n + 3) = 2.$$

Therefore, the GCD is either 1 or 2. But $2n + 3$ is odd, so the GCD cannot be 2. Hence

$$\boxed{\gcd(2n + 3, 4n + 8) = 1}.$$

Example 20

Determine

$$\gcd(n^2 + n + 1, n + 2).$$

We can rewrite the quadratic expression as

$$n^2 + n + 1 = (n - 1)(n + 2) + 3.$$

Therefore,

$$\gcd(n^2 + n + 1, n + 2) = \gcd(n + 2, 3).$$

Thus, the GCD is either 1 or 3, depending on n .

Recovering Hidden Numbers

The identity

$$\gcd(a, b) \operatorname{lcm}(a, b) = ab$$

allows us to recover information about two numbers.

Example 21

Two positive integers have GCD 18 and LCM 1260. One of the integers is 180. Find the other.

Let the other integer be b . Then

$$18 \cdot 1260 = 180b.$$

Therefore,

$$b = \frac{18 \cdot 1260}{180} = 126.$$

Checking,

$$\gcd(180, 126) = 18$$

and

$$\operatorname{lcm}(180, 126) = 1260.$$

Thus, the other number is

$$\boxed{126}.$$

Remark N3.1.22. The product identity alone does not always guarantee that the information is possible.

Suppose someone claims that two numbers have GCD 12, LCM 630, and one number is 72. The product identity gives

$$b = \frac{12 \cdot 630}{72} = 105.$$

But

$$\gcd(72, 105) = 3,$$

not 12. Therefore, no such pair exists.

Example 23

Find all unordered pairs $\{a, b\}$ of positive integers satisfying

$$\gcd(a, b) = 12$$

and

$$\text{lcm}(a, b) = 360.$$

Write

$$a = 12x, \quad b = 12y.$$

Since 12 is the entire GCD,

$$\gcd(x, y) = 1.$$

Using the product identity,

$$(12x)(12y) = 12 \cdot 360,$$

so

$$xy = 30.$$

The relatively prime factor pairs of 30 are

$$(1, 30), \quad (2, 15), \quad (3, 10), \quad (5, 6).$$

Therefore, the unordered pairs $\{a, b\}$ are

$$\boxed{\{12, 360\}, \{24, 180\}, \{36, 120\}, \{60, 72\}}.$$

Remainder Problems

Example 24 (Same Remainder)

Find the least positive integer greater than 5 that leaves remainder 5 when divided by 12, 18, and 30.

Then

$$n - 5$$

is divisible by each of 12, 18, and 30. Therefore,

$$\text{lcm}(12, 18, 30) \mid n - 5.$$

Since

$$\text{lcm}(12, 18, 30) = 180,$$

we have

$$n = 180k + 5$$

for some nonnegative integer k . The least such integer greater than 5 is

$$\boxed{185}.$$

Example 25 (One Less Than a Multiple)

Find the least positive integer n that leaves remainder 1 when divided by 2, remainder 2 when divided by 3, remainder 3 when divided by 4, and remainder 4 when divided by 5.

In every case, n is one less than a multiple of the divisor. Thus,

$$n + 1$$

is divisible by 2, 3, 4, and 5. Therefore,

$$n + 1 = \text{lcm}(2, 3, 4, 5) = 60,$$

so

$$\boxed{n = 59}.$$

Geometric Applications

Lattice Points on a Diagonal

Consider the line segment joining $(0, 0)$ to (a, b) , where a and b are positive integers.

Claim N3.1.26 — The number of lattice points on the segment, including both endpoints, is

$$\boxed{\text{gcd}(a, b) + 1}.$$

Proof. Let

$$d = \text{gcd}(a, b).$$

Write

$$a = dx, \quad b = dy,$$

where $\text{gcd}(x, y) = 1$.

The segment can be divided into d equal steps, each moving x units horizontally and y units vertically. Therefore, the lattice points are

$$(0, 0), (x, y), (2x, 2y), \dots, (dx, dy).$$

There are

$$d + 1$$

such points. □

Example 27

How many lattice points lie on the segment from $(0, 0)$ to $(18, 30)$?

Since

$$\text{gcd}(18, 30) = 6,$$

the number of lattice points is

$$\boxed{7}.$$

How Many Squares Does a Diagonal Cross?

Suppose a rectangle is divided into an $m \times n$ grid of unit squares. A diagonal is drawn from one corner to the opposite corner.

Claim N3.1.28 — The number of unit squares crossed by the diagonal is

$$m + n - \gcd(m, n).$$

Proof. As the diagonal travels across the rectangle, it crosses:

- $m - 1$ interior vertical grid lines;
- $n - 1$ interior horizontal grid lines.

If no crossings occurred at grid intersections, the diagonal would enter

$$1 + (m - 1) + (n - 1) = m + n - 1$$

squares (think: why do we need the 1+?).

However, whenever the diagonal passes through an interior lattice point, it crosses a vertical and horizontal line simultaneously. Such a point was counted twice but should increase the number of squares by only 1.

The diagonal contains

$$\gcd(m, n) + 1$$

lattice points including the two endpoints, so it contains

$$\gcd(m, n) - 1$$

interior lattice points.

Therefore, the number of squares crossed is

$$m + n - 1 - (\gcd(m, n) - 1) = m + n - \gcd(m, n).$$

□

Example 29

How many unit squares does the diagonal of an 18×30 rectangle cross?

$$18 + 30 - \gcd(18, 30) = 48 - 6 = 42.$$

Three-Dimensional Extension

A line segment passes from one corner of an $a \times b \times c$ rectangular box to the opposite corner.

Claim N3.1.30 — The number of unit cubes it passes through is

$$a + b + c - \gcd(a, b) - \gcd(a, c) - \gcd(b, c) + \gcd(a, b, c).$$

This formula resembles inclusion–exclusion:

- begin by counting crossings of each family of coordinate planes;
- subtract crossings that occur simultaneously along grid lines;
- add back crossings that occur at lattice points.

This result is considerably harder, so it is left as an exercise.

True or False?

Determine whether each statement is true or false. If it is false, provide a counterexample.

1. Consecutive odd positive integers are relatively prime.
2. If $a \mid b$, then $\gcd(a, b) = a$.
3. If $a \mid b$, then $\text{lcm}(a, b) = b$.
4. If $\gcd(a, b) = d$, then $d \mid (a + b)$.
5. If $\gcd(a, b) = 1$, then $\gcd(a, a + b) = 1$.
6. The GCD of two positive integers always divides their LCM.

Recognizing GCD and LCM Problems

Keywords can be helpful, but the safest method is to translate the problem into divisibility statements.

Problems that often use the GCD involve:

- the greatest number of identical groups;
- the largest possible square tile;
- equal spacing with no remainder;
- etc.

In these problems, the answer usually must *divide* several given quantities.

Problems that often use the LCM involve:

- when repeating events next occur together;
- the smallest quantity divisible by several numbers;
- the least number of objects that can be grouped in several ways;
- common remainders;
- etc.

In these problems, the answer usually must be *divisible by* several given quantities.

Example 31 (Quick Classification)

Determine whether each problem is most naturally solved using a GCD or an LCM.

1. Several ribbons are cut into the longest possible equal pieces.
2. Three buses return to a station at different regular intervals.
3. A floor is covered with the largest possible congruent square tiles.
4. Find the smallest number divisible by 12, 18, and 25.
5. Divide several types of objects into the greatest possible number of identical packages.